

[illegible]

Diagram illustrating three cascaded amplifiers, each represented by a yellow box labeled "Amplifier", "Amplifier1", and "Amplifier2". Each amplifier has a green input signal labeled "SG_1", "SG_2", and "SG_3" respectively, entering from the left. Below each amplifier box, the file path "File: amplify.kicad_sch" is indicated.

The diagram shows the MA702GQ-P IC (IC1) with the following connections:

- SSCK:** Connected to pin 17 (MGH_17) via a 10k resistor (R8) to GND. A label MGH_MGH is present.
- VDD:** Connected to pin 13 (VDD) via a 3V3 supply and a 1uF capacitor (C13) to GND.
- SSD:** Connected to pin 1 (SSD) via a 10k resistor (R10) to GND. A label MGH_SPI1_SCK is present.
- MG_A2:** Connected to pin 2 (MG_A2) via a 10k resistor (R11) to GND. A label MGH_MGL is present.
- MG_Z3:** Connected to pin 4 (MG_Z3) via a 10k resistor (R12) to GND. A label MGH_SPI1_MOSI is present.
- MISO:** Connected to pin 7 (MISO) via a 10k resistor (R14) to GND. A label MGH_SPI1_MISO is present.
- MG_B:** Connected to pin 6 (MG_B) to GND. A label MGH_SPI1_NSS is present.
- MG_PWM:** Connected to pin 9 (MG_PWM) via a 10k resistor (R13) to GND. A label MGH_PWM is present.
- Other pins:** GND.2 (pin 18), MGH (pin 16), SCK (pin 15), NC (pin 14), and VDD (pin 13) are also shown.

The image displays three schematic diagrams of phase blocks, labeled Phase_A, Phase_B, and Phase_C. Each block is represented by a yellow rectangle with a red border. The connections and labels are as follows:

- Phase_A:**
 - Inputs: HIN_A → HIN, LIN_A → LIN.
 - Outputs: VISENSE ← VISENSE_A, VSENSE ← VSENSE_A, VPH ← VPH_A.
 - Connection: VPH_A is connected to J1 Conn_A via a red line with a '1'.
 - File: phase.kicad_sch
- Phase_B:**
 - Inputs: HIN_B → HIN, LIN_B → LIN.
 - Outputs: VISENSE ← VISENSE_B, VSENSE ← VSENSE_B, VPH ← VPH_B.
 - Connection: VPH_B is connected to J2 Conn_B via a red line with a '1'.
 - File: phase.kicad_sch
- Phase_C:**
 - Inputs: HIN_C → HIN, LIN_C → LIN.
 - Outputs: VISENSE ← VISENSE_C, VSENSE ← VSENSE_C, VPH ← VPH_C.
 - Connection: VPH_C is connected to J3 Conn_C via a red line with a '1'.
 - File: phase.kicad_sch

The schematic diagram illustrates the internal connections of the USB-C to UART module. It features a central USB4510-03-1-A IC (J7) and a USBLC6-25C6 IC (U2).

USB Connection: The USB4510-03-1-A IC is connected to the USB-C port (J4) via USB_DBCC2 (R15, 5k1). The IC pins are labeled: A1, A4, A5, A6, A7, A8, A9, A12, B5, B6, B7, B8, MH1, MH2, MH3, MH4, CC2, DP2, DN2, SBU2, DN1, SB1, VBUS_2, GND_2, GND_1, VBUS_1, CC1, DP1, DN1, SB1, VBUS_1, GND_1. The USB4510-03-1-A IC is also connected to the USB_DBCC1 pin of the USB-C port (J4) via a 1uF capacitor (C19) and a 5k1 resistor (R16).

Bluetooth header UART: The USB4510-03-1-A IC is connected to the Bluetooth header (J5) via a 3V3 supply and a 100nF capacitor (C58). The header pins are labeled: 1 (TXD), 2 (RXD), 3 (GND), 4 (GND), 5 (GND), 6 (GND).

Power Connector: The USB4510-03-1-A IC is connected to the Power Connector (J6) via a 3V3 supply and a 100nF capacitor (C58). The connector pins are labeled: 1 (VCC), 2 (GND), 3 (GND), 4 (GND), 5 (GND), 6 (GND).

TAG SWD Connector: The USB4510-03-1-A IC is connected to the TAG SWD Connector (J6) via a 3V3 supply and a 100nF capacitor (C58). The connector pins are labeled: 1 (VCC), 2 (RESET), 3 (MCU_N_RST), 4 (MCU_SWCLK), 5 (MCU_SWCLK), 6 (MCU_SWCLK).

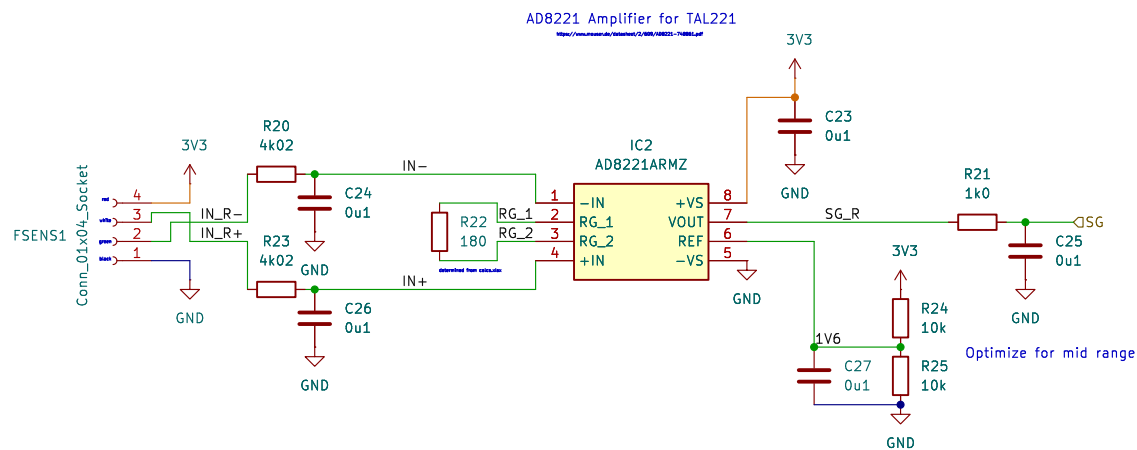
Other Components: The schematic includes a 3V3 supply, a 100nF capacitor (C58), a 5k1 resistor (R15), and a 1uF capacitor (C19).

Y1
CS08B91-24M

PFO_OSC_IN 4
1 XTAL_2
XTAL_1 GND_1

PF1_OSC_OUT 3
2 C21 6p8
GND

C22 6p8
GND



Sheet: /Amplifier1/
 File: amplify.kicad_sch

Title:

Size: A4

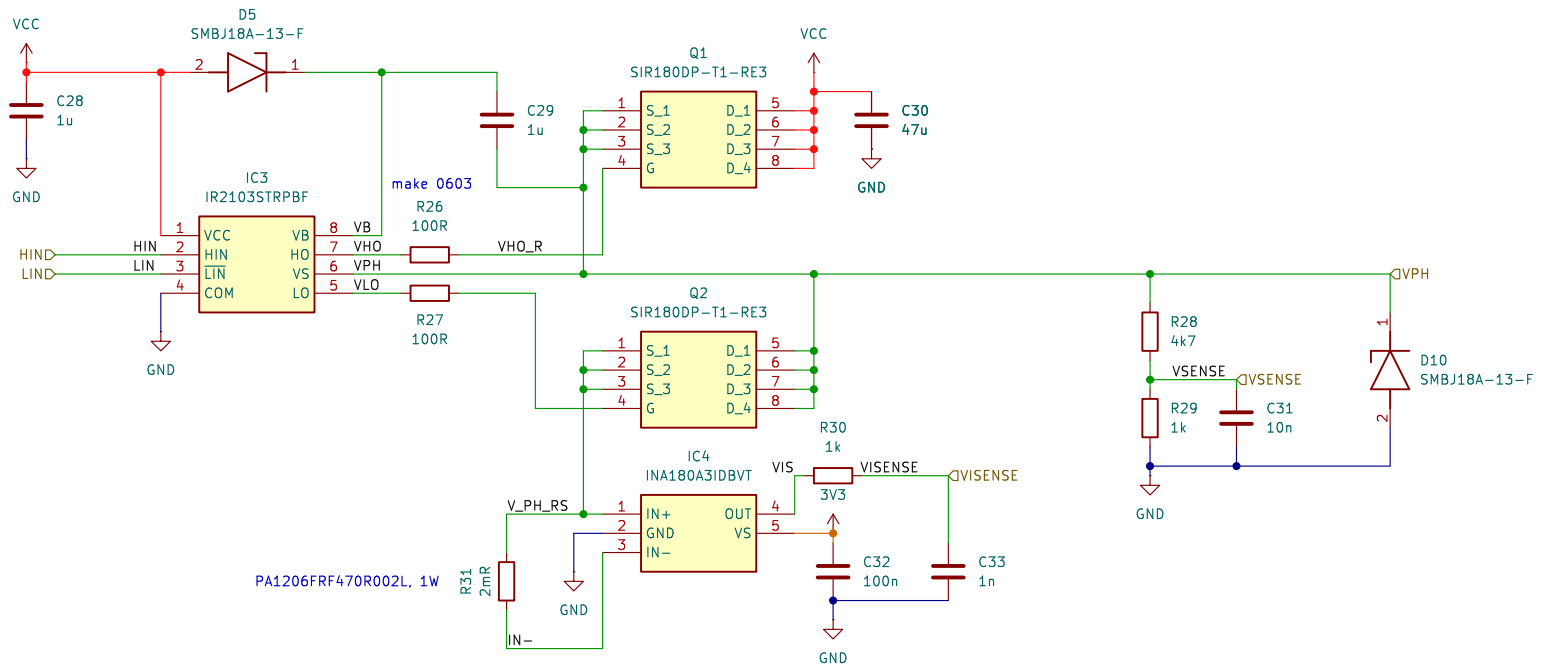
Date:

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Rev:

Id: 2/7

PHASE



Amplifier to measure voltage dro

Sheet: /Phase_A/
File: phase.kicad_sch

Title:

Size: A4

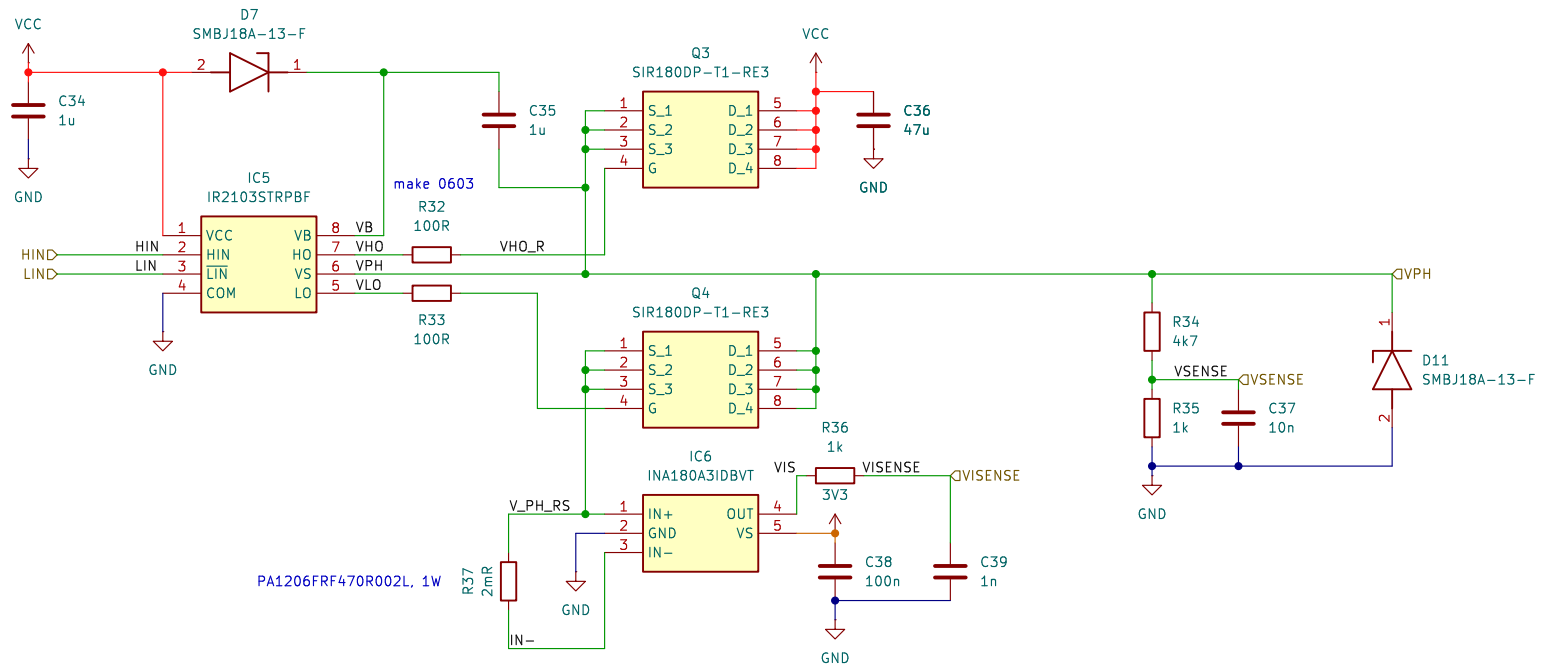
Date _____

Rev:

KiCad E.D.A. 9.0.1

Id: 3/7

PHASE



Amplifier to measure voltage drop

Sheet: /Phase_B/
File: phase.kicad_sch

Title:

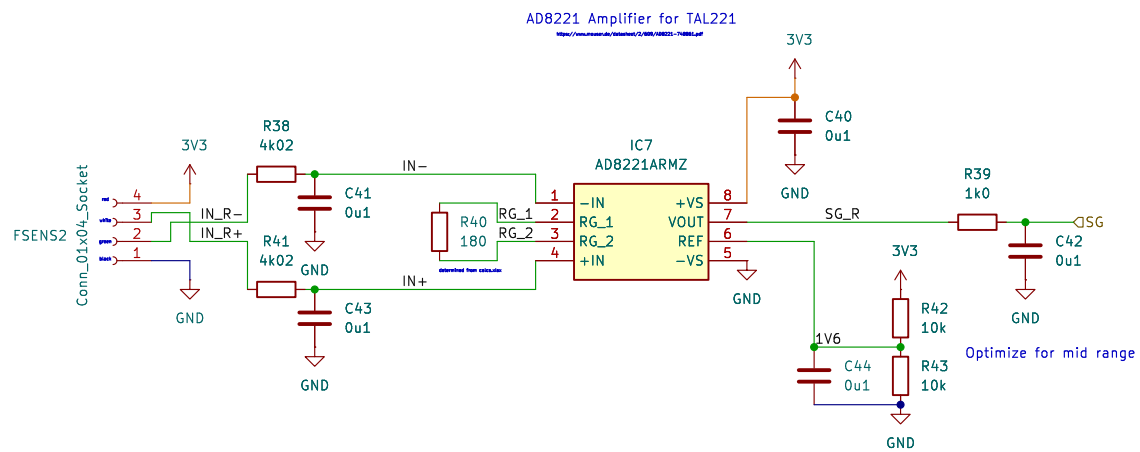
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Date:

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Rev:

Id: 4/7



Sheet: /Amplifier/
File: amplify.kicad_sch

Title:

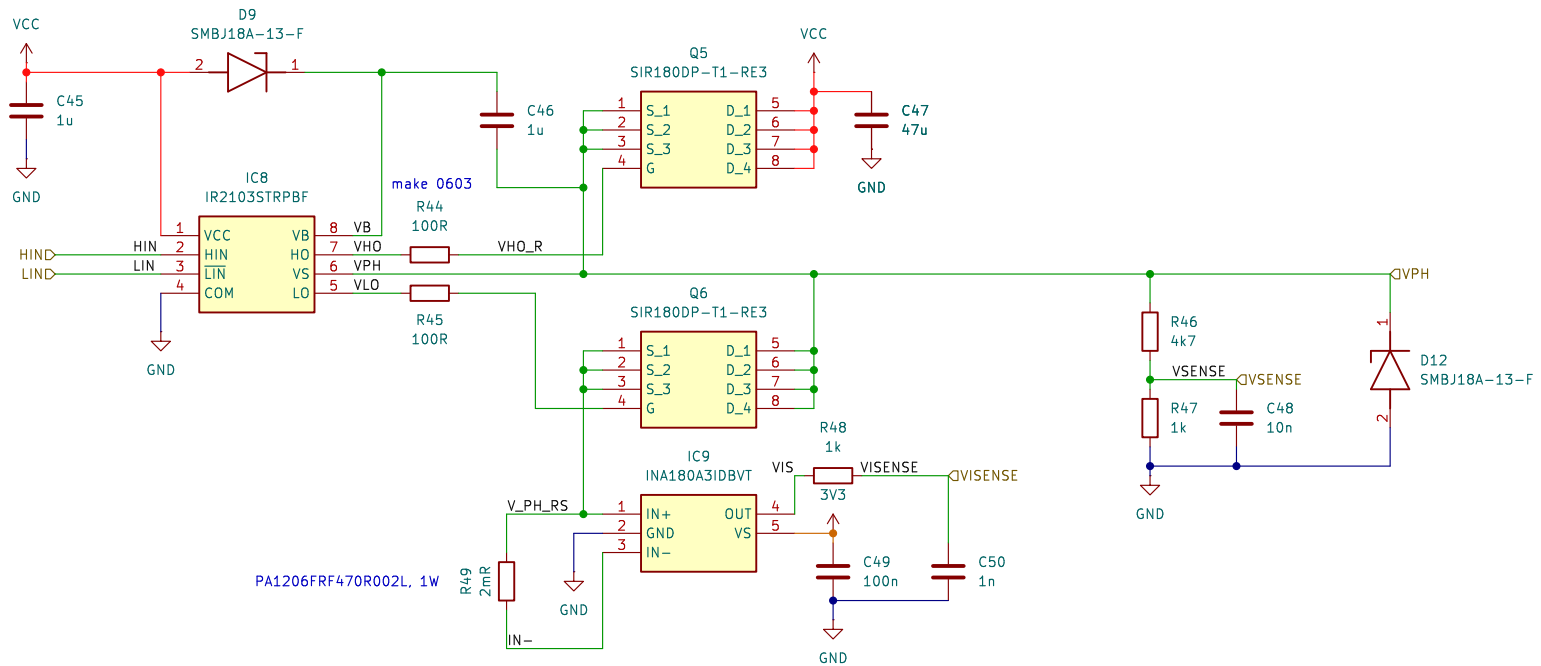
Size: A4 Date:

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Rev:

Id: 5/7

PHASE



Amplifier to measure voltage dro

Sheet: /Phase_C/
File: phase.kicad_sch

Title:

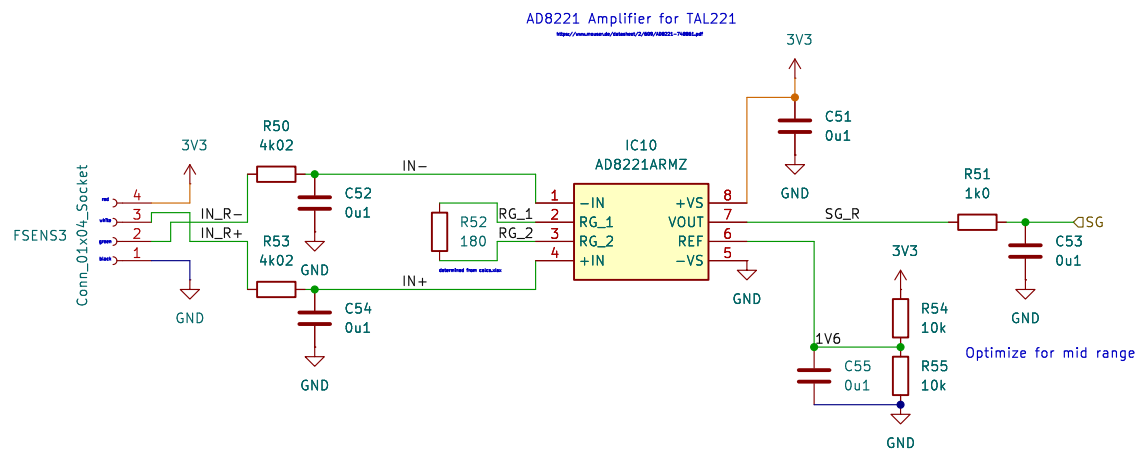
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Id: 5/7



Sheet: /Amplifier2/
 File: amplify.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 9.0.1

Rev:

Id: 6/7